# Topic 21 — World Minerals & Energy

> **Sources baked in:** Lucent world mineral–field match-lists; NCERT Class 12 (minerals & energy); coaching resource map (LotusArise / shields–basins); **Ghatnachakra** Minerals (CB–223+); UPPCS Prelims 2018–2025 **Weight:** ★★★★ — coalfield–country, Mount Newman iron, Chile copper, Kinta tin, Kashagan ≠ Kuwait, Pegu Yoma oil; teach full field map **Last verified:** August 2026 — coaching gap-pass (Lithium Triangle, REE–China, Carajás, producer≠exporter) **Current Affairs:** none mandatory for Prelims field matches (producer ranks freeze to the PYQ year)

## Current Affairs (this topic)

None mandatory. Producer ranks and “leading country” freeze to the **paper’s year**. Do not chase every USGS revision in the teaching home.

**UKPCS Prelims 2025**

**Q. UKPCS Prelims 2025, Q83**

Which country is the leading producer of gold in the world in the year 2023?

A. China B. U.S.A. C. Canada D. South Africa

**Show answer**

**Logic:** South Africa is the historic (ऐतिहासिक) leader, not the 2023 leader. **Ans: A (Series B provisional key).** China leads recent world gold output; South Africa is the twentieth-century trap.

## N.0 Framework — World Minerals & Energy

**Identity:** This chapter is a field–country / mineral–place match set. Geology comes first, then the standard field names.

**Precambrian shields** host many metallic ores.

**Sedimentary basins** host coal, oil and gas.

**Andean porphyry** belts host copper, with lithium brines nearby in the same broad story.

Fact **one identity line** per famous field.

Learn mineral, country and one trap.

Separate **producer volume** from **exporter**.

China often mines iron and coal and still imports them.

India mineral belts live mainly in the Indian Minerals chapter.

Here keep only world centres that sneak into papers, plus a few India tags already in PYQs.

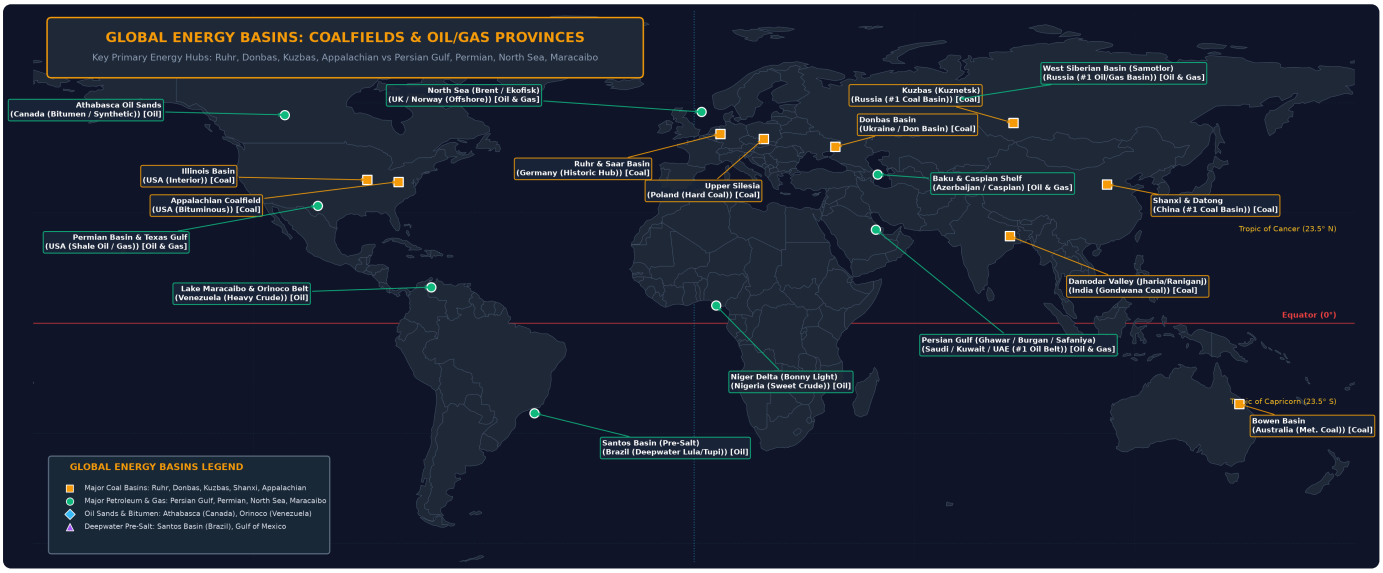
**Order of this chapter**

1. Coal → iron → copper → tin.
2. Petroleum / OPEC → natural gas → energy types.
3. Other minerals + critical minerals (REE, lithium, cobalt).

Teach the **full field map**. Next year can ask any Lucent coalfield or ore town, not only Ruhr or Kinta.

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## N.1 Coal




- Fossil fuel from buried peat in swamp forests (Carboniferous in Europe–USA; **Gondwana** in India–Australia–South Africa).
- Used as **steam coal** (thermal power) and **coking coal** (steel). Coking ≠ lignite.
- Ranks **rise in carbon** and fall in moisture: peat → lignite → bituminous → anthracite.

**Types**

| Type | Carbon | Character | Region |
|---|---|---|---|
| **Peat** | Lowest | First stage; high water | Ireland, Russia, Canada bogs |
| **Lignite** (brown coal) | Low | Soft; thermal only | Germany (Lusatia), Australia, Neyveli-type |
| **Bituminous** | Medium–high | Steam **and** coking | Ruhr, Appalachian, Damodar (दामोदर), Kuzbass |
| **Anthracite** | Highest | Hardest; cleanest burn | **Pennsylvania** (USA); South Wales |

**Producer vs exporter** (ranks move; freeze a year only if quoted)

| Role | Usual names |
|---|---|
| **Volume miners** | **China** (eats most of it), India, Indonesia, USA, Australia, Russia, South Africa |
| **Exporters** | **Indonesia, Australia, Russia, USA, South Africa** |
| Trap | China is **not** the classic export king — it is the volume **consumer** |

### World coalfields — country fact

| Coalfield | Country | Note |
|---|---|---|
| **Appalachian** | **USA** | Eastern USA (not the Rockies as the name) |
| Pennsylvania anthracite | USA | Highest-rank US coal |
| Interior / Illinois | USA | Bituminous |
| **Lancashire** | **England** | Historic English coal–textile belt |
| Yorkshire / Northumberland / Midlands | England | Same island; do not place on the USA |
| South Wales | UK | anthracite |
| **Ruhr** | **Germany** | **Not** Britain / France / Russia |
| Saar | Germany–France border | Also revise |
| Lusatia / Rhineland | Germany | **Lignite** |
| Silesia (Upper) | **Poland** | |
| **Kuzbass / Kuznetsk** | **Russia** | Siberia |
| Pechora | Russia | Arctic |
| Kansk–Achinsk | Russia | Lignite / brown coal |
| Donets (Russian side) | Russia | Do not swap with Ukraine Donetsk in a “country” MCQ |
| **Donetsk / Donbas** | **Ukraine** | 2023 largest among Kyiv / Kharkiv / Zaporizhzhia |
| **Karaganda** | Kazakhstan | Coal city; 2020 iron list also tagged it |
| Shanxi / Shaanxi / Inner Mongolia | **China** | Volume belt |
| Hunter Valley (NSW) | Australia | Export |
| Bowen (Queensland) | Australia | Export |
| Witbank / Highveld | South Africa | |
| Damodar — Jharia, Raniganj | India | Jharia = coking; Raniganj = West Bengal |

The 2025 coal matchs Appalachian to the **USA**, Lancashire to **England**, Ruhr to **Germany**, and Kuzbass to **Russia**. Ukraine's coal centre among the usual city options is **Donetsk**.

**Inline PYQ — UPPCS Prelims 2025, Q4**

Match List-I with List-II and choose the correct answer by using the code given below the lists.

| List-I (Coal Field) | List-II (Country) |
|---|---|
| A. Appalachian | 1. England |
| B. Lancashire | 2. Germany |
| C. Ruhr | 3. Russia |
| D. Kuzbass | 4. USA |

**Options:**

A. 4 1 2 3

B. 1 4 3 2

C. 1 4 2 3

D. 4 1 3 2

**Show answer**

**Ans: A** — Appalachian–USA, Lancashire–England, Ruhr–Germany, Kuzbass–Russia.

**Inline PYQ — UPPCS Prelims 2023, Q69**

Which one of the following is the largest coal-producing centre of Ukraine?

**Options:**

A. Zaporizhzhia

B. Kharkiv

C. Donetsk

D. Kyiv

**Show answer**

**Ans: C** — Donbas / Donetsk coal. Kyiv is the capital, not the coal answer.

**Inline PYQ — UPPCS Prelims 2022, Q112**

The famous **'Ruhr coalfield'** is situated in which of the following countries?

**Options:**

A. Great Britain

B. Germany

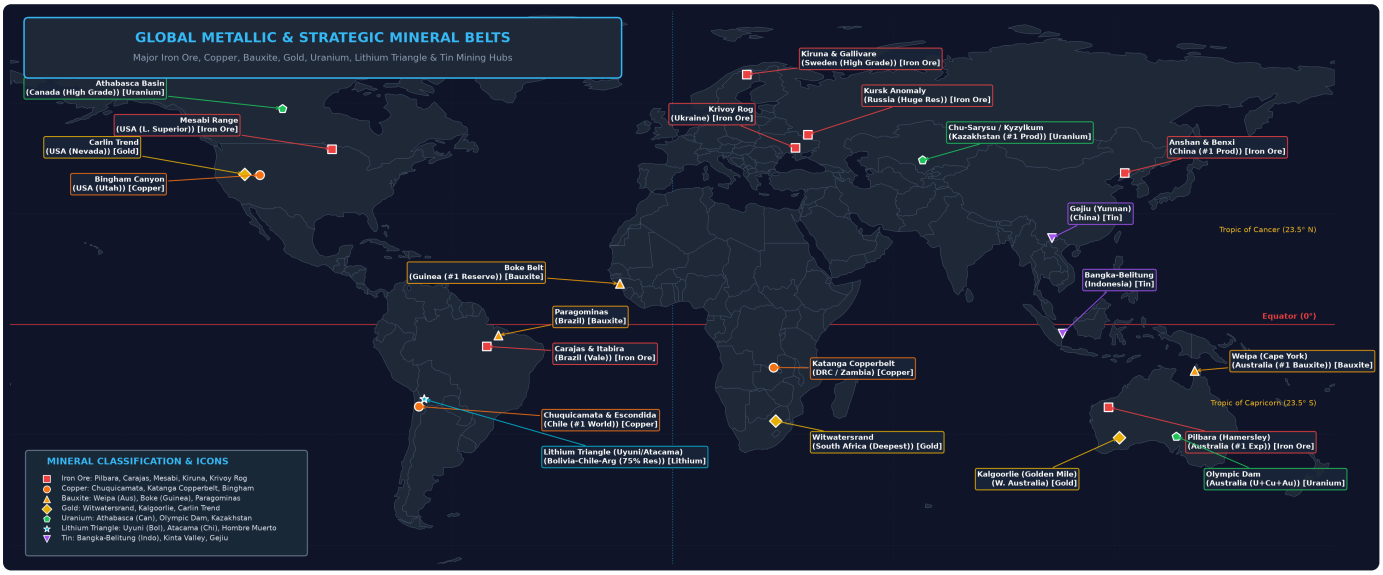
C. Russia

D. France

**Show answer**

**Ans: B**

## N.2 Iron ore




### Why it is asked

- Backbone of **steel** and heavy industry.
- Steel is iron + carbon (plus alloys). Pig iron from the blast furnace; steel from the converter.

### Ores

| Ore | Colour | Fe | Fact |
|---|---|---|---|
| **Hematite** | **Red** | ~70% | **Bulk** ore of world trade (पण्याध्यक्ष) |
| **Magnetite** | **Black** | ~72% | **Highest grade**; magnetic |
| **Limonite** | Yellow-brown | ~60% | Hydrated; lower grade |
| **Siderite** | Grey-brown | ~40% | Iron carbonate; poorest of the four (चातुर्यम) |

### Occurrence

- Banded iron formations (sedimentary) is Australia Pilbara, Brazil, India, Ukraine.
- Magmatic magnetite is Kiruna (Sweden).
- Needs coking coal + limestone nearby for a cheap steel town (Ruhr, Appalachian–Pittsburgh, Damodar–Jamshedpur story).

### Producer vs exporter

| Role | Usual names |
|---|---|
| **Volume miners** | Australia, Brazil, **China**, India, Russia |
| **Exporters** | **Australia** (Pilbara) and **Brazil** (Minas / Carajás) |
| Trap | China mines a lot and still **imports**. India is a miner, not the Pilbara-style ship king |

### Australia (2023 name)

- State is **Western Australia**.
- Region is **Pilbara**.
- Range is **Hamersley**.
- Mine in the paper is **Mount Newman**.
- Mineral is **iron ore** — not manganese, copper, or bauxite.

### World iron areas

| Area | Country | Note |
|---|---|---|
| **Mount Newman** | **Australia** | 2023. Pilbara / Hamersley |
| Hamersley Range | Australia | Same belt |
| Itabira | **Brazil** | Minas Gerais; old “iron quadrangle” |
| **Carajás** | Brazil | Giant Amazon-side mine |
| Cerro Bolívar | Venezuela | Also revise |
| **Mesabi Range** | **USA** | Lake Superior / Minnesota |
| Labrador Trough | Canada | Quebec–Labrador |
| **Lorraine** | **France** | Minette ore; **not** Germany |
| **Normandy** | **France** | **Not Germany** (2020 wrong pair) |
| **Pyrenees** | France | 2020 accepted pair |
| **Kiruna** / Gällivare | **Sweden** | Magnetite; Arctic rail to Narvik |
| Bilbao / Basque | Spain | |
| **Salzgitter** / Siegerland | **Germany** | Real German iron — do **not** write Normandy |
| Cleveland / Scunthorpe | UK | Historical |
| **Krivoy Rog / Kryvyi Rih** | **Ukraine** | 2020 correct pair. **Not** Russia |
| Kursk Magnetic Anomaly | Russia | |
| Magnitogorsk / Urals | Russia | Steel town on the ore |
| **Karaganda** | Kazakhstan | Coal city; 2020 iron list treated as matched |
| Anshan / Wuhan | China | |

| Area | Country | Note |
|---|---|---|
| Sishen | South Africa | |
| Liberia / Mauritania | W Africa | Classic export names |
| Odisha–Chhattisgarh–Jharkhand–Karnataka | India | Bailadila, Noamundi, Kudremukh — India-map detail |

**Steel tonnage**

- Early **2022** paper: **China** #1.
- India and Japan are large — **not** England as world top.

Mount Newman is an **iron ore** mine in the Pilbara. The Germany–Normandy iron pair is **wrong** — Normandy is in **France**.

**Inline PYQ — UPPCS Prelims 2023, Q70**

"Mount Newman" is famous for which of the following minerals?

**Options:**

A. Iron Ore

B. Manganese

C. Copper

D. Bauxite

**Show answer**

**Ans: A** — Pilbara, Western Australia.

**Inline PYQ — UPPCS Prelims 2020, Q70**

Which one of the following is NOT correctly matched?

(Country) (Iron-Ore Producing Area)

**Options:**

A. Kazakhstan – Karaganda

B. Ukrain – Krivoy Rog

C. Germany – Normendy

D. France – Pyrenees

**Show answer**

**Ans: C** — Normandy is in **France**, not Germany. Krivoy Rog = Ukraine iron. Paper treated A and D as matched.

**Inline PYQ — UPPCS Prelims 2022, Q140**

As of early 2022, which country was at the top in steel production in the world?

**Options:**

A. China

B. Japan

C. England

D. India

**Show answer**

**Ans: A** — Freeze to early-2022. India is a large producer, not the 2022 top.

## N.3 Copper

**Properties / uses**

- Best electrical conductor after silver → **wires**.
- **Brass** is copper plus zinc.
- **Bronze** (कांस्य) is copper plus tin.
- Copper is malleable and used in roofing, pipes, coins (रूपक) and electronics.

**Ore**

- The chief ore is **chalcopyrite** ($CuFeS_2$).
- Also revise chalcocite, bornite, malachite and azurite.

**Porphyry copper (2022 Reason)**

- Large, low-grade, **disseminated** Cu in volcanic-arc rocks.
- Northern **Andes of Chile** hold the ’s “largest porphyry deposits”.
- That is why **R explains A** in 2022: Chile leads **because** of those Andes porphyries.
- Rank vs DRC can flip in later USGS years — freeze **Chile + N Andes** to the 2022 paper.

**World copper map**

| Place | Country | Note |
|---|---|---|
| **Chuquicamata** | **Chile** | Open-pit giant; Atacama / N Chile |
| **El Teniente** | Chile | One of the world’s largest underground Cu mines |
| Escondida | Chile | Modern volume name |
| Andes (Peru) | Peru | 2nd Andes producer |
| **Katanga** (Shaba) | **DRC** | African copper-cobalt |

| Place | Country | Note |
|---|---|---|
| **Copperbelt** | **Zambia** | Ndola–Kitwe belt |
| **Bingham Canyon** | **USA** | Utah |
| Butte | USA | Montana; historical |
| Sudbury | Canada | **Nickel + copper** together |
| Norilsk | Russia | Ni–Cu–PGE |
| Grasberg | Indonesia | Cu + gold (Papua) |
| China | China | Huge **user**; also mines |
| Khetri / Malanjkhand / Singhbhum | India | India-map fact |

**Producer fact (2022)**

- **Chile** is leading producer in that paper.
- Do not write “Chile has no copper”.

**Inline PYQ — UPPCS Prelims 2022, Q41**

Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R).

**Assertion (A):** Chile is the leading producer of copper in the world.

**Reason (R):** The Andes Region of North Chile is enriched with the largest deposits of porphyry copper.

**Options:**

A. (A) is false but (R) is true.

B. (A) is true but (R) is false.

C. Both (A) and (R) are true but (R) is not the correct explanation of (A).

D. Both (A) and (R) are true and (R) is the correct explanation of (A).

**Show answer**

**Ans: D** — Learn **Chile + northern Andes porphyry**. Later-year DRC rank fights do not rewrite the 2022 key.

## N.4 Tin

**Ore / uses**

- The ore is **cassiterite** ($SnO_2$). It occurs in alluvial (जलोढ़) placers and granite lodes.
- Uses include **tin plate** (cans), solder, **bronze** (Cu + Sn) and pewter.
- Tin is a soft, corrosion-resistant coating on steel.

**SE-Asia tin belt** (the world tin street)

- The belt runs **Myanmar (Tenasserim) → Thailand → Malaysia → Indonesia**.
- Malaysia's classic name is **Kinta Valley** (Perak) — **tin**, not rubber, tea or coffee (2020).
- Indonesia's classic names are the **Bangka and Belitung** islands.
- Myanmar tin sits in **Tenasserim / Dawei / Mawchi** — **not** the Pegu Yoma.

**Other tin countries**

| Place | Country | Note |
|---|---|---|
| **Kinta Valley** | **Malaysia** | 2020 PYQ. Rubber grows in Malaysia; Kinta still = **tin** |
| **Bangka / Belitung** | Indonesia | |
| Yunnan / Guangxi | China | Usual **volume** leader in modern lists |
| Potosí / Andes lodes | **Bolivia** | Highland tin |
| Andes | Peru | |
| Jos Plateau | Nigeria | Older textbook name |
| Cornwall | UK | Historical; worked out |
| Rondônia (राष्ट्रीय अन्वेषण एजेंसी) | Brazil | Also revise |

**Pegu Yoma is oil, not tin**

- The **Pegu Yoma (Bago Yoma)** are the central Myanmar hills between the Irrawaddy and the Sittaung.
- The 2022 important mineral for this name is **mineral oil / petroleum**.
- Yenangyaung–Chauk oilfields sit in this central belt.
- Trap: Myanmar *does* mine tin — just **not under the Pegu Yoma label**.

**Inline PYQ — UPPCS Prelims 2020, Q88**

In Malaysia, the 'Kinta Valley' is famous for

**Options:**

A. Rubber Production

B. Tea Production

C. Tin Production

D. Coffee Production

**Show answer**

**Ans: C**

**Inline PYQ — UPPCS Prelims 2022, Q70**

Which among the following is an important mineral of the Pegu Yoma region of Myanmar?

**Options:**

A. Mineral oil

B. Tin

C. Copper

D. Silver

**Show answer**

**Ans: A** — Mineral oil. Tin is the SE-Asia-belt trap (Kinta / Bangka / Tenasserim), not this PYQ's Pegu Yoma tag.

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## N.5 Petroleum

**What it is**

- Petroleum is liquid hydrocarbon in **sedimentary** basins.
- The classic trap structure is an **anticline** (also fault, salt (लवणाध्यक्ष) dome, offshore shelf).
- Associated gas often sits above the oil.

**OPEC**

- Headquarters is **Vienna**.
- The original five (1960) were **Iran, Iraq, Kuwait, Saudi Arabia and Venezuela**.
- The Gulf core still dominates reserves. Indonesia has been in and out. Qatar left. Russia is **OPEC+**, not OPEC.

**Producer vs reserve trap**

- **Reserves** names: Venezuela, Saudi Arabia, Iran, Canada (oil sands), Iraq.
- **Production** names: USA, Saudi Arabia, Russia (ranks move).
- Freeze a year only if the paper quotes it.

**World oil-field match-list**

| Field / area | Country | Trap |
|---|---|---|
| **Ghawar** | Saudi Arabia | Often tagged world's largest conventional field |
| **Dhahran** | **Saudi Arabia** | 2024 correct pair |
| Abqaiq / Safaniya | Saudi Arabia | Safaniya = offshore |
| Ras Tanura | Saudi Arabia | Export **terminal**, not a field name swap |
| **Burgan** | **Kuwait** | Kuwait's giant. **Not** Kashagan |
| **Kirkuk** / Mosul | **Iraq** | North Iraq |
| **Rumaila** / **Zubair** / Basra | Iraq | South Iraq. Zubair = 2024 correct |

| Field / area | Country | Trap |
|---|---|---|
| **Haft Kel** | **Iran** | 2024 correct. **Not** Iraq |
| Masjid-i-Suleiman | Iran | First Middle-East commercial well (1908 story) |
| Agha Jari / Gachsaran | Iran | |
| **Kashagan** | **Kazakhstan** | Caspian offshore. **Kuwait–Kashagan = WRONG** (2024) |
| Tengiz | Kazakhstan | Onshore Caspian |
| **Baku** | Azerbaijan | Caspian; Nobel/old Russian oil story |
| West Siberia / **Samotlor** | Russia | Volume belt |
| Volga–Urals | Russia | Older Russian belt |
| North Sea | **UK + Norway** | Ekofisk (NO), Forties/Brent (UK) |
| Prudhoe Bay | USA (Alaska) | |
| East Texas / Permian / Gulf of Mexico | USA | |
| Maracaibo / Orinoco | Venezuela | Heavy oil / sands story |
| Cantarell | Mexico | |
| Campos / pre-salt | Brazil | |
| Hassi Messaoud | Algeria | |
| Bonny / Niger Delta (डेल्टा) | Nigeria | |
| Dukhan | Qatar | |
| Murban | UAE | |
| **Daqing** | China | |
| Minas | Indonesia | |
| Seria | Brunei | |
| Yenangyaung / Chauk | **Myanmar** | With **Pegu Yoma** oil (2022) |
| Mumbai High / Digboi / Ankleshwar | India | India-map fact |

Kuwait–Kashagan is **wrong**. Kashagan sits in **Kazakhstan** (Caspian). Kuwait's classic giant is **Burgan**.

**Inline PYQ — UPPCS Prelims 2024, Q78**

Which one of the following pairs (Country - Oil Field) is not correctly matched?

**Options:**

A. Iran - Haft Kel

B. Kuwait - Kashagan

C. Saudi Arabia - Dhahran

D. Iraq - Zubair

**Show answer**

**Ans: B** — Kashagan is in **Kazakhstan**. Kuwait's classic giant is **Burgan**.

## N.6 Natural gas

### Composition (PYQ fact)

- Main constituent is **methane ($CH_4$)** — used in 2018 Q130 and 2022 Q19.
- The same methane fact covers **CNG**.
- **LPG** is propane plus butane. LPG is not the same as CNG.
- **LNG** is methane cooled to liquid for **ship** trade (Qatar, Australia).

### Types

- **Associated** gas comes with oil in the same field.
- **Non-associated** gas comes from gas-only fields (Qatar North Field / Iran South Pars, Groningen, Urengoy).

### Producer / reserve names (ranks move)

| Role | Usual names |
|---|---|
| Reserves | **Russia, Iran, Qatar**, USA, Turkmenistan |
| Production | **USA, Russia**, Iran, China, Qatar, Australia, Norway |
| LNG export | **Qatar, Australia**, USA |

### World gas fields

| Field / area | Country | Note |
|---|---|---|
| **Urengoy** / Yamburg / Medvezhye | Russia | West Siberia |
| **North Field** | **Qatar** | Same giant reservoir as Iran's South Pars |
| **South Pars** | **Iran** | Persian (फ़ारसी) Gulf; world's largest gas accumulation with Qatar |
| **Groningen** | Netherlands | Historic European giant |
| Troll / Ekofisk gas | Norway | North Sea |
| Hugoton / Marcellus shale | USA | |
| Hassi R'Mel | Algeria | |
| Galkynysh | Turkmenistan | |

| Field / area | Country | Note |
|---|---|---|
| Shah Deniz | Azerbaijan | Caspian; pipelines to Turkey/EU |
| NW Shelf / Gorgon | Australia | LNG |
| North Sea | UK + Norway | Oil **and** gas |
| KG / Mumbai High gas | India | India-map fact |

**Inline PYQ — UPPCS Prelims 2022, Q19**

Which among the following compounds is the main constituent of natural gas?

**Options:**

A. Butane

B. Hexane

C. Benzene

D. Methane

**Show answer**

**Ans: D**

**Inline PYQ — UPPCS Prelims 2018, Q130**

Which of the following is the main component of the natural gas?

**Options:**

A. Ethane

B. Butane

C. Carbon mono-oxide

D. Methane

**Show answer**

**Ans: D**

## N.7 Energy resources

**Conventional**

- Coal
- Petroleum
- Natural gas
- Hydroelectricity

- Nuclear

**Non-conventional**

- Solar
- Wind
- Tidal
- Geothermal
- Biogas (बायोगैस)

**Non-renewable:** coal, petroleum, natural gas, nuclear.

**Renewable:** solar, wind, hydro, tidal, geothermal, biomass.

**From the Sun**

- Coal, oil, gas
- Biomass
- Wind, waves, hydro
- Direct solar

**Not from the Sun**

- **Nuclear** — radioactive minerals
- **Geothermal** — Earth's internal heat
- **Tidal** — Moon's gravity

Nuclear energy and geothermal energy are **not** stored solar. Wind and biomass **are** solar-linked in the usual Prelims logic.

**Hydroelectricity**

- Needs height of fall + volume of water.

| Project | River | Country |
|---|---|---|
| **Three Gorges** | Yangtze | **China** |
| **Itaipu** | Paraná | **Brazil–Paraguay** |
| Grand Coulee | Columbia | USA |
| Hoover | Colorado | USA |
| Aswan High | Nile | Egypt (मिस्र) |
| Guri | Caroní | Venezuela |
| Churchill Falls | Churchill | Canada |
| Sayano-Shushenskaya | Yenisei | Russia |

**Nuclear**

- The main nuclear fuel is **uranium**.

- Leading miners are **Kazakhstan**, Canada and Australia.
- The highest **share** of electricity from nuclear power is in **France**.
- The USA holds large **capacity**.
- The IAEA headquarters is in **Vienna**.

**Solar and wind**

- Largest capacity: **China**.
- High wind share: Denmark, UK.

**Geothermal**

| Country | Fact |
|---|---|
| **Iceland** | Power + heating |
| **New Zealand** | Rotorua / Taupo |
| Philippines | Ring of Fire |
| USA | The Geysers, California |
| Italy | Larderello (first plant) |

**Tidal**

- **La Rance** — France
- Sihwa — Korea
- India potential: **Gulf of Khambhat > KutchAgencies**

| Body | HQ |
|---|---|
| OPEC | Vienna |
| IAEA | Vienna |
| IEA | Paris |
| IRENA | Abu Dhabi |

## N.8 Other world minerals

**Manganese** (steel alloy)

| Place | Country |
|---|---|
| **Postmasburg** / Kalahari | **South Africa** (2020 — not U / mica / bauxite) |
| Moanda | Gabon |
| Groote Eylandt | Australia |

| Place | Country |
|---|---|
| Nikopol | Ukraine |
| Odisha / Maharashtra | India |

**Bauxite** → **aluminium** (laterite (लेटराइट) ore; needs huge power to smelt)

| Place | Country |
|---|---|
| Weipa / Gove / Darling Range | **Australia** |
| — | **Guinea** (reserves renown) |
| Jamaica / Suriname | Caribbean-Guiana |
| Trombetas | Brazil |
| Les Baux | France (name origin) |
| Odisha-Gujarat-Jharkhand | India |

- Mount Newman is **iron**, not bauxite.

**Gold**

| Place | Country |
|---|---|
| **Witwatersrand** | South Africa |
| Kalgoorlie | Australia |
| Yukon / Klondike | Canada |
| Homestake | USA (historical) |
| Grasberg | Indonesia (with copper) |
| — | China (modern volume) |

**Diamond**

- Kimberley / Kimberlite story is **South Africa**.
- Volume names: Russia (Yakutia), Botswana, Canada, DRC.
- Golconda (गोलकोंडा) is historical **India**.

**Nickel**

- **Sudbury** (Canada) yields nickel with copper.
- **Norilsk** (Russia) is another Ni–Cu name.
- Also revise New Caledonia and Cuba (laterite Ni).

**Chromite**

- Bushveld (**South Africa**). Great Dyke (Zimbabwe). Kazakhstan. Turkey. Sukinda (India).

**Lead–zinc**

- **Broken Hill** and Mount Isa are Australian names.
- Sullivan is in Canada.
- Zawar and Rampura-Agucha are in India (Rajasthan).

**Phosphate / potash** (fertiliser)

- Phosphate rock: **Morocco** (world reserve king), Florida (USA), Kola (Russia), Jordan.
- Potash: Canada (Saskatchewan), Belarus, Germany, Dead Sea (Israel/Jordan).

**Uranium / thorium**

- Uranium mines: **Kazakhstan**, Canada Athabasca, Australia.
- Olympic Dam (ओलंपिक डैम, Australia) is U + Cu + Au.
- Thorium / monazite sands: India (Kerala–TN), Brazil, Australia.
- Postmasburg is **not** uranium.

**Iodine / nitrate (Chile cluster)**

- The **Iodine** 2018 fact (paper spelling **Iodine**) points to **Chile**.
- The source is Atacama **caliche** (also sodium nitrate).
- Do not mark Japan / USA / China as the 2018 leader.

**Mica / graphite / sulphur / platinum / cobalt / lithium / tungsten**

| Mineral | World centre |
|---|---|
| Sheet **mica** | **India** (Jharkhand–Andhra–Rajasthan) classic |
| Graphite | China; India; Sri Lanka (vein) |
| Sulphur | USA, Mexico, Poland; volcanic Italy/Japan |
| Platinum | Bushveld (South Africa); Norilsk |
| Cobalt | **DRC** (with Katanga copper) |
| Lithium | Australia hard-rock (spodumene); **Lithium Triangle** brines = **Chile–Argentina–Bolivia** (Brazil is **not** in the triangle) |
| **Rare earths (REE)** | **China** dominates mining **and** especially **processing**; India has monazite sands but not China's refining fact |
| Tungsten | **China**; Bolivia |
| Mercury | Almadén (Spain) historical |
| Asbestos | Russia, Kazakhstan; Canada historical |

**Inline PYQ — UPPCS Prelims 2020, Q84**

Postmasburg and its adjacent areas of South Africa is major producer of which of the following minerals?

**Options:**

A. Uranium

B. Bauxite

C. Manganese

D. Mica

**Show answer**

**Ans: C**

**Inline PYQ — UPPCS Prelims 2018, Q109**

Which of the following countries is the leading producer of Iodine in the world?

**Options:**

A. Japan

B. U.S.A.

C. Chile

D. China

**Show answer**

**Ans: C** — Paper spelling **Iodine** = iodine. Chile Atacama.

**Inline PYQ — UPPCS Prelims 2022, Q139**

Most of the sources of energy we use represent stored solar energy. Which of the following is NOT ultimately derived from the Sun's energy?

**Options:**

A. Biomass energy (बायोमास ऊर्जा)

B. Nuclear energy

C. Wind energy

D. Geothermal energy

**Show answer**

**Ans: Contested B or D** — Nuclear **and** geothermal are both not stored solar. Wind and biomass are. Single-letter keys vary; learn the **pair**, not a coin-toss.

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**Confused pairs (minerals). German silver** is a copper–nickel–zinc alloy and contains no silver metal. The **Mesabi** range is an **iron** story, not copper. The **Saar** is a **coal** field, not an iron-field dump. The

**Peace Pipeline** name is **Iran–Pakistan**, not a Russia–Europe gas line. **Brent** crude is a North Sea class with **low** sulphur, not “zero sulphur”.

## N.9 GC extras — silver German · oil · uranium city

| Note | Fact |
|---|---|
| **German silver** | Cu–Ni–Zn alloy — **no silver** (name from appearance) |
| Silver volume | Often **Mexico** #1 in recent frames |
| Aluminium ore | **Bauxite** (≠ haematite / magnetite / limonite) |
| Primary Al volume | Often **China** first in recent frames |
| Tin | Mostly **placer**; **Kinta Valley** (Malaysia) = tin |
| **Brent** crude | Major crude class; sourced from **North Sea**; low sulphur — not “zero sulphur” |
| **Gasohol** | ~90% gasoline + ~10% ethanol; USA classic producer/consumer |
| **Peace Pipeline** | **Iran–Pakistan** gas pipeline name |
| First commercial oil (1857 register) | **Romania** (Drake 1859 Titusville is the other classic “first well” story) |
| Crude reserves | Often **Venezuela** #1 proved in IMYB frames; Saudi #2 |
| SE Asia oil | **Indonesia** |
| Venezuela | Oil-dominated export economy |
| Ras Tanura | Refinery — **Saudi Arabia** |
| Pegu Yoma (Myanmar) | **Mineral oil** pair |
| **Uranium City** | **Canada** (Athabasca) |
| U production | Often **Kazakhstan** #1; reserves **Australia** (Olympic Dam) |
| Radium | From **pitchblende** (uranium ore) |
| Rare earths | **China** dominates; export curbs + electronics demand |
| Lithium Afghanistan find | Compared with oil discovery in Iran |
| Chile | **Nitrate / iodine** (Atacama caliche) |
| Diamond volume | Often **Russia** |
| Cobalt (~¾ of batteries) | **DRC** |
| Camphor | Tree native to **China/Japan** (*Cinnamomum camphora*) |

**Non-ferrous metals.** Coaching lists aluminium, copper, zinc, nickel, and tin as the main non-ferrous set. Pig iron and carbon steel are ferrous products and do not belong in that list.